

Electronic commerce comes to the 'Net'

J A Edelhelt and M R Miller

While the fundamental business-to-business (e.g. supply chain) trading model has not significantly changed in hundreds, if not thousands, of years, many of the underlying business processes and trading practices have changed and continue to evolve. In some cases, these changes have been more successful than others.

While the primary purpose to this paper is to discuss the effect of Internet-based electronic commerce on supply chain business processes and trading practices, it also provides a historical review of electronic commerce. As George Santayana once said: "Those who cannot remember the past are condemned to repeat it."

1. Introduction

What is electronic commerce (EC)? Depending on whom you ask, it may mean using a credit card to buy something from a World Wide Web (Web) site. Another person may say that EC is electronic data interchange (EDI). A third person may say that EC supports all aspects of business trading, including the sending and receiving of electronically signed documents.

The High Level Strategy Group for ICT (HLSG) [1] defined EC as a set of tools and services which:

- reduce the cost of creating, moving, managing and processing the documents on which are built relationships between trading partners,
- improve business entities processes by reducing the cost of operating business and improving overall quality,
- increase revenue by opening new market channels.

1.1 Electronic commerce functionality

A fully functional EC system must minimally support three functions:

- interactive transactions,
- database transactions,
- secure messaging.

Interactive transactions refer to a human/computer interaction. Today, this interaction usually uses a Web browser interface. The browser interface may be used for selecting items from an electronic catalogue or for completing a form. The form may be used for interacting with a database or application. An example of a forms-based application could be a more user-friendly interface to an EDI system.

Database transactions refer to computer/computer interaction. Traditionally, this approach could be viewed as two computer systems passing commercial transactions between themselves. These transactions frequently were predefined, formatted EDI messages that were usually transmitted over an X.400 or specialist EDI value-added network.

The third functional area is secure messaging. Secure messaging utilises security services such as privacy, integrity, authentication, and non-repudiation. Many of these security services would be provided by an electronic notary.

1.2 Supply chain business model

The supply chain business model has not significantly changed in hundreds, if not thousands, of years. Figure 1 presents a simplified version of this model.

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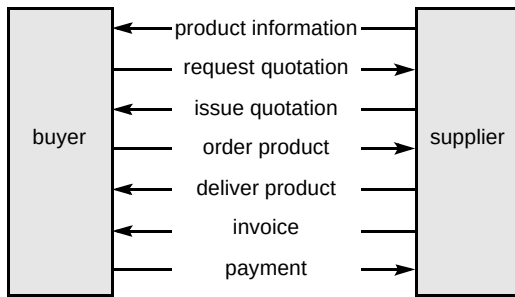


Fig 1 Supply chain information flows.

What has changed, however, is the way this information is exchanged. In the beginning, this information was exchanged face-to-face. This exchange may have been verbal or may have even been a non-verbal pantomime. As society and cultures evolved, this exchange may have taken on a written form. In any case, this exchange usually was on a one-to-one basis. With the development of the 'information age', more advanced methods of exchanging information have evolved — and this evolution continues.

1.3 Scope of this paper

This paper focuses on supply chain (business-to-business) electronic trading and does not address retail (consumer-to-merchant) electronic commerce [2]. While this paper may mention areas such as payment services, trust services, or encryption, detailed discussions of these topics may be found elsewhere [3 — 5].

2. A historical perspective

2.1 Manual information processing

Paper-based trading systems have existed for thousands of years and remain in widespread use. Over time, assurance mechanisms in terms of authentication of origin, uniqueness, resistance to alteration, etc, were developed to support these trading systems. These measures have been enshrined in trade law by the legal systems of all countries. Today, equivalent assurances and legal cover are demanded for trade with modern electronic systems.

To support international trade where the buyer is often unknown to seller, complex structures of agents, international treaties, recognition of the seals of public notaries, and the like have grown up over the years. In some countries, national laws still prohibit the use of documentation derived from electronic systems such as EDI and require fully notarised paper from the point of origin.

A first move away from total reliance on paper being sent by physical mail was made in the 1950s with the adoption of Telex as a relatively high-speed delivery medium. Telex even achieved legal acceptability as trade documentation due to the fact that it was then considered

quite difficult to forge the telecommunications network information embedded in the messages. However, the use of electronic transmission of the messages for part of the journey still meant that paper records existed at all stages in the creation of a trade transaction and these required manual activity to progress, complete and file.

The arrival of the facsimile has done little to advance things. It has improved availability, since it uses the ordinary phone network, and increased the speed of message transmission. This increase in speed has whetted the appetite for more responsive supply chains. While facsimile allows hand-written documents to be transmitted, these documents may be readily altered by computer, and, therefore, are inherently untrustworthy. Lastly, the wide use of thermal paper also poses problems of message archiving.

In more recent years, the development of faster international transport, coupled with much improved telecommunications has meant a vast increase in trade activity that is simply not supportable with manual processing of paper.

2.2 Stand-alone IT information processing

As computers became cheaper and more widely available, most large businesses began to rely upon them for internal processing of orders, invoices and accounts. The introduction of IT to the business process allowed sophisticated companies greater control of inventory and other costs, with consequent competitive advantage.

The main characteristic of computers was their ability to produce immense volumes of paper very quickly, which all needed physical transmission to customers, suppliers and so forth. Studies have indicated that a high proportion of all computer input is the output of some other computer, plus a liberal sprinkling of errors, retyped by an army of bored clerical staff. To ensure that errors were eliminated, a second team of people would duplicate the activity and the result printed out and posted back to the originator for them to check against their own computer output. This huge waste of time and effort led a number of organisations to experiment with direct computer-to-computer communication. This experimentation led to the development of an EDI system.

2.3 EDI

2.3.1 Overview

EDI is now widely used by business to support the supply chain and is supported by accepted international

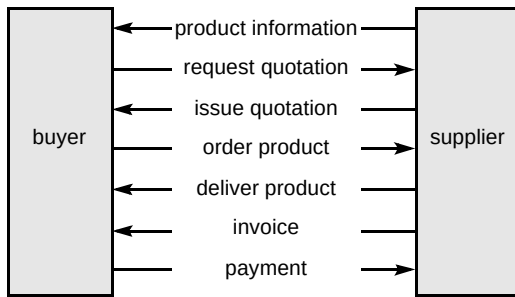


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2.3.1 Overview

EDI is now widely used by business to support the supply chain and is supported by accepted international

standards. This section will provide background to the next stage, where EDI is expected to become integrated with the Internet in some way.

2.3.2 Requirements

Successful implementation of electronic commerce requires that the parties have confidence that their messages are being exchanged with the right partner and that their messages are reaching the destination without alteration. In some sensitive cases, there are additional assurances needed that the messages have not been observed by third parties. They also need to be certain what the legal position will be in case of any dispute.

In the case of EDI, each pair of partners establishes a trading agreement. This agreement sets out the rules within the context of an existing supply contract, the technical standards and the network services to be used. The EU has produced a recommended model for such agreements [6]. Because a contractual situation exists (e.g. a pre-arranged trading agreement), the parties can trade in complete confidence, which is less easily assured with simple Internet trading.

A key component in EDI has been the value added network (VAN), such as BT's EDI*Net. The VANs operate a store and collect mail service for all its subscribers, allowing them to send multiple messages efficiently in a single telecommunications session. There is the added benefit that the VAN offers a 24-hour service, holding messages until the recipient is ready to retrieve them, eliminating the need for everyone to be on-line all the time.

2.3.3 Sample transaction/information flow

The EDI transaction flows generally follow those of the equivalent paper process (Fig 2).

Most EDI activity is in support of bulk supply contracts where the prices are pre-agreed, but the message standards allow the option of a quotation stage as well. As the order is transmitted electronically, it is unnecessary to use an acknowledgement to prove that it was rekeyed correctly. Rather, the acknowledgement message is used only where it is not possible to supply within the agreed time-scales. Likewise, the option of an electronic 'delivery note' is often waived since the customer will soon be aware of failure to supply when the production line stops! Such process simplifications, possible because of electronic commerce, often generate greater savings than would be expected from the simple saving of a rekeying operation.

One relatively recent enhancement is that some banks offer a related service so that when a customer company settles several invoices with a single electronic payment, a remittance advice identifying the invoices being paid, is simultaneously transmitted with the payment. Unfortunately, this approach has generally not been accepted by the business community. This lack of acceptance may be due to the relatively higher cost of this approach as compared to the simpler Bankers Automated Clearing System (BACS) transfers.

2.3.4 EDI standards

For EDI to be successful in a whole industry sector there is a need for standards. Obviously there are considerable gains to be made if every retail supermarket can use a

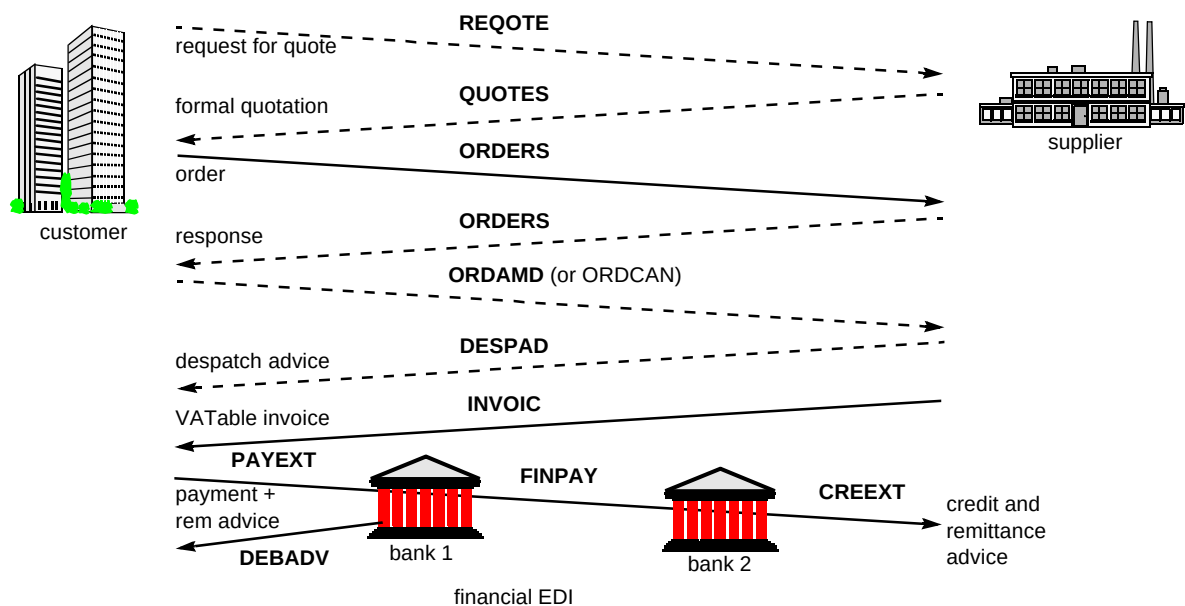


Fig 2 Ordering transaction sequence illustrated by EDIFACT messages.

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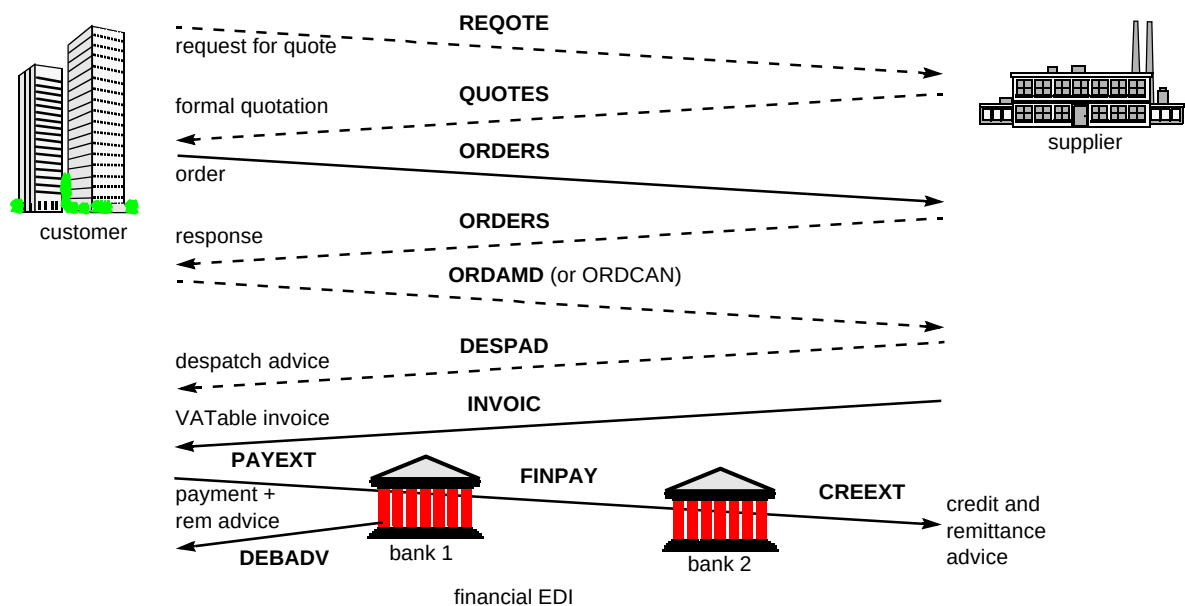


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common set of messages to address what is often a small group of common suppliers, such as branded foodstuffs. This standardisation has taken place under the guidance of the relevant national trade associations.

A key feature of EDI is that it links databases in a structure-independent manner. Unlike direct database access techniques, such as SQL where the actual field structure must be known to the enquirer, individual parties use an interface package (which may well use SQL) between their database and the outside world. Only the information that is needed for the particular message is extracted from the database, then it is placed into the standard message envelope. Recipients in turn take the fields from the incoming message and place them in their own database (which may be differently structured) and process them within the database application. This process secures private information, provides a standard data structure, and isolates application development activity in one partner company from affecting their trading partners.

National standards — the early development of EDI took place independently in the major industrialised countries and this led to different national message standards. For example, in the UK a number of companies agreed to use EDI and developed message standards known as 'TRADACOM', which are administered by the UK Article Numbering Association (ANA). Similarly, in the USA the ANSI X.12 EDI standards have become almost universal, while across Europe the motor manufacturing industry has adopted the ODETTE standards.

EDIFACT — many organisations recognised that the emerging 'EDI Tower of Babel' was a barrier to international trade. Given its international status, the United Nations (UN) established the UN/EDIFACT [7] standards and regional message-definition bodies were created to support message development and ratification. While message development is time-consuming, the result is that a single message set may be used by a company for trading with its global partners.

One aspect of using global EDIFACT message standards is that message sets and sequences have to be devised to support various national and international legal requirements. At a lower level, each message (e.g. invoice) defines data elements for virtually every eventuality and most trading activities are able to leave many of them empty. Specific industry sectors have thus adopted sub-sets of EDIFACT known as 'trade conventions.' These conventions identify which messages will be used and which fields are required; but, the fact remains that EDI messages, which have taken some 15 years to develop, reflect the requirements of real business processes and information flows. For this reason, the data flows and content should ideally be reused by any alternative electronic trading environment such as the Internet.

2.4 X.400 messaging

The PTTs, in planning to replace the Telex service, defined the X.400 messaging system. The X.400 system supports binary attachments to text messages, guarantees delivery, and provides automatic acknowledgements.

Two notable features, not currently available on the Internet, are message authentication and non-repudiation. Message authentication is a security service that assures the identity of the message's sender and receiver. Non-repudiation provides independent assurance that identifies when the recipient initially opened (e.g. read) the message.

These features make X.400 very attractive for electronic commerce. Unfortunately, although the X.435 sub-section of X.400 provides for the transmission of EDI messages such as EDIFACT, many other formatted documents have been designed as proprietary 'standards' for specific industries and applications.

2.5 The Internet

With the arrival of the World Wide Web, attention has been focused on the Internet. What began as a playground for the academic community has become a household word. The Web's ability to transmit multimedia documents has allowed the Internet to become the new advertising and purchasing mechanism for products and services.

2.5.1 Internet and EDI coexistence

The worlds of Internet and EDI currently possess quite different characteristics. While typical EDI users each have computerised operations with interconnected databases sharing details of orders, invoices, etc, the typical Internet user has a home or office PC which interacts with a database elsewhere. This is illustrated in Fig 3.

These current differences arise chiefly from the way in which electronic commerce is used by companies and individuals.

The Internet browser mechanism used for Web ordering has largely been developed to satisfy the needs of individuals rather than corporations. It is a simple matter for individual users to decide whether or not to purchase, and then (subject to the above-mentioned concerns about transaction security) to submit an electronic form detailing

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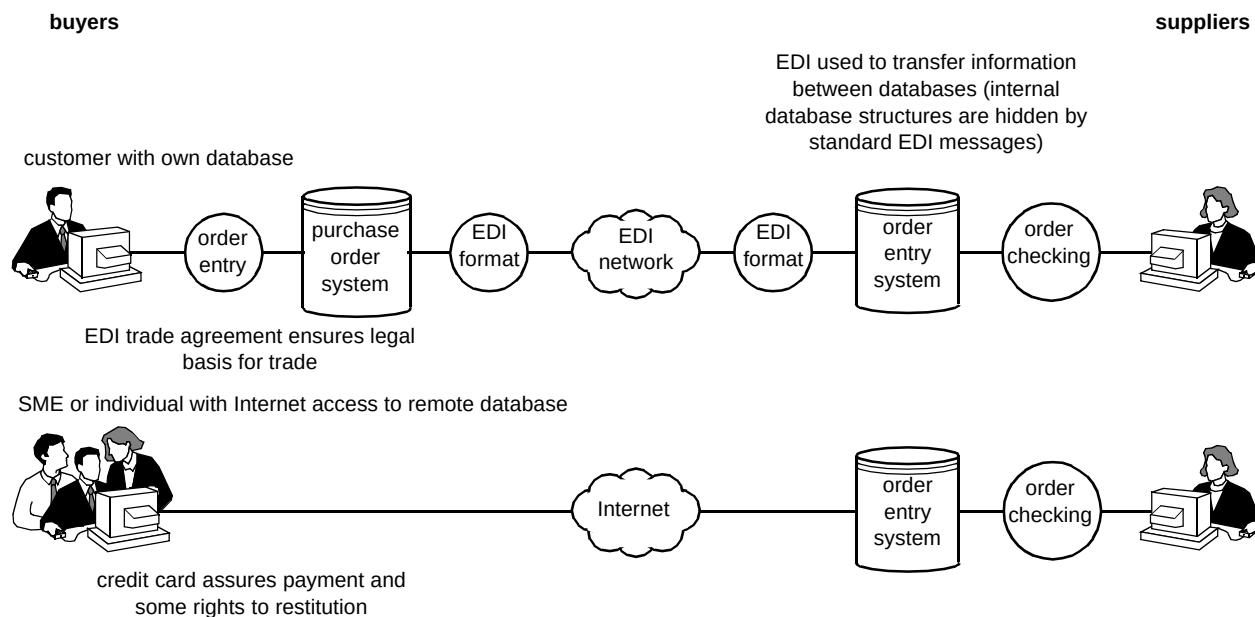


Fig 3 One database or two?

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By contrast, generally accepted accounting principles require that the ordering process be subject to controls and checks. Some of these controls and checks include that the appropriate authorisation was granted and that all transaction details are entered in the financial systems or databases. Thus, the Web ordering mechanisms must be modified to link into corporate databases at the buyer's end, including processes to obtain internal authorisation.

While these two worlds do not yet interact to a significant extent, it is inevitable that they will converge as technology develops. The Bank of America and Lawrence Livermore National Laboratories implemented a CommerceNet pilot project that transmitted 'classic' EDI messages in ANSI X.12 format over the Internet [8]. Attention is also being focused on the transmission of the information content of 'classic' EDI using Internet protocols such as SMTP with a MIME attachment. Others are investigating the definition of 'business objects' [9] to carry the transaction information. It is to be hoped that this will utilise the existing business definitions of transaction flows embodied in EDI, which can itself be represented as an object model [10].

2.5.2 On-line catalogues

What is an on-line catalogue? One simplistic view is to take an existing paper catalogue and place it in an electronic medium. This electronic medium might use a stand-alone or networked CD-ROM, a proprietary on-line service, or an

open, Internet-based environment. The first 'on-line' catalogues were CD-ROM based. One early adopter of CD-ROM catalogues was the electronic parts industry. Historically, electronic parts distributors published large paper catalogues on a semi-annual basis. These paper catalogues frequently exceeded 400 or 500 pages and usually provided product specifications and recommended retail or standard industry prices. While the paper catalogues offered multiple indices (e.g. by product type, by manufacturer), paper catalogues were unwieldy to use and expensive to produce and distribute. The advent of the CD-ROM catalogue allowed the electronic parts distributor to produce catalogues more frequently, at a lower production and distribution cost. While these CD-ROM catalogues frequently provided search engines that were much more functional than static indices, order placing was still a manual process. This process required the user to call the vendor for a quotation or to place an order. Over time, CD-ROM catalogues were enhanced to allow the user to complete and print an order form that could then be faxed to the vendor. One interesting extension to this environment was the EDI-CAT system. Developed by Syntegra, the BT systems integration organisation, EDI-CAT allowed the buyer to select products from a CD-ROM catalogue, but to then place an order using EDI*Net, BT's X.400-based EDI network. While CD-ROM-based catalogues achieved some cost savings, buyers were still saddled with an environment that offered static information and required significant human interaction.

Over the last several years, several organisations have experimented with proprietary services that download a customised electronic catalogue to the user's personal computer (PC). This approach has several advantages over the CD-ROM based catalogues. These advantages include

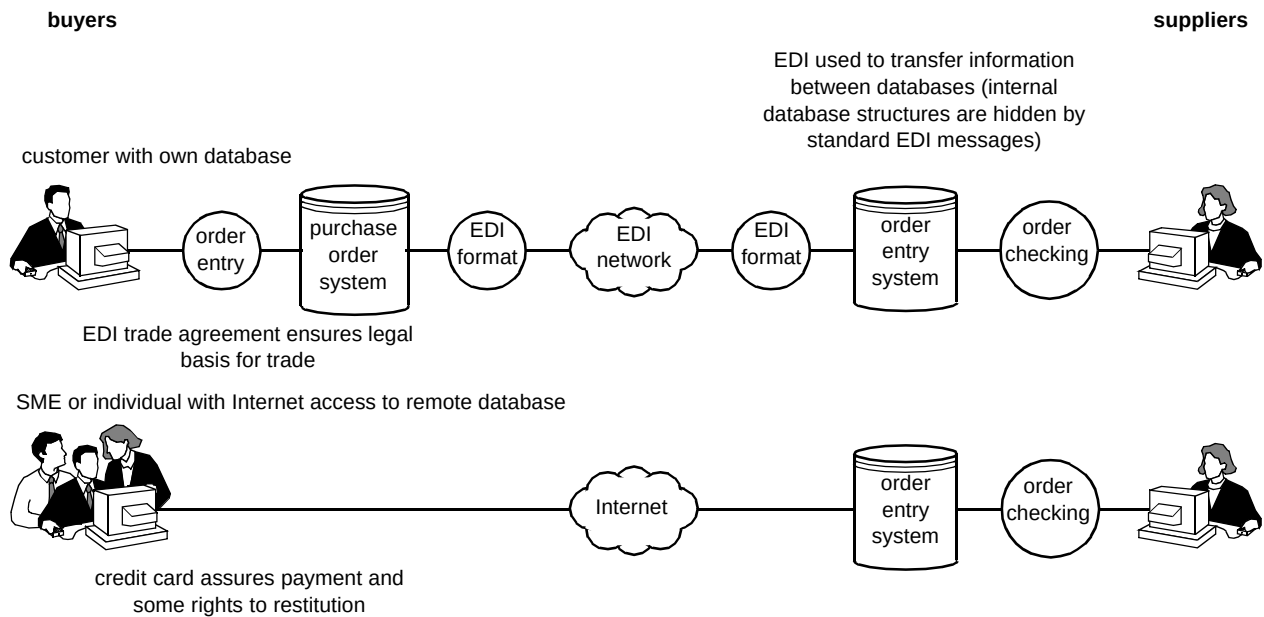


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more frequent product updates or additions, and may include customer-specific prices or products. As mentioned above, these catalogues tend to use closed, proprietary technology and do not provide an electronic ordering option like EDI-CAT. Rather, the user can select products, create, print, and send an order by facsimile to the vendor.

Dynamic, Internet-based catalogues have recently begun to appear. These catalogues use open, standards-based Web browsers for access. Based on predefined user profiles, these dynamic catalogues may be configured to present customer-specific information such as special products (e.g. private brands), prices, discounts or promotions.

2.5.3 Intelligent agents

What is an intelligent agent? Intelligent agents are much more than the search engines currently found in electronic catalogues. Evolving intelligent agents technology will allow users to concurrently search heterogeneous catalogues, identify products, negotiate pricing and delivery terms, and place orders. Intelligent agents may also identify alternate sources or products.

Many organisations, including BT, are currently performing intelligent agent research [11]. Over time, more and more intelligent agent technology will be included in Internet-based electronic commerce products and environments.

3. Internet electronic commerce issues

Perhaps a more significant issue is a basic difference in the current EDI and Internet trading models. Specifically, the EDI trading model is based on a pre-arranged trading agreements, while the Internet model currently assumes that trading occurs on an *ad hoc* basis.

3.1 Support for pre-existing and ad hoc trading

EC requires more than a technological foundation to succeed. Specifically, there are a number of legal issues and trading practices that need to be addressed. These issues and practices include the following.

- What proof must buyers provide to sellers that they are indeed whom they claim and that they are authorised to obligate their organisation?
- How can the buyer or seller be assured that the transaction is valid and binding?

- What assurance must the seller provide that the pricing displayed is valid and has not been modified or altered?

Generally speaking, the above issues and practices are not a concern if an EDI-like pre-existing trading agreement is in place. In such cases, the pre-existing trading agreement is a legally binding document that defines what is a contractually enforceable transaction. However, this last statement may not be true for *ad hoc* trading environments. While identification, authentication, non-repudiation, and integrity security services (e.g. electronic signatures and encryption) may offer some technological solutions, no consistent legal basis exists that defines what an electronic signature is and when it is acceptable. Likewise, whereas the UK and other European governments have recognised the need for licensing trust service providers [4] (e.g. trusted third parties), the same cannot be said for the United States.

3.2 Removal of manual processes and interfaces

While electronic catalogues and intelligent agents are EC enablers, much more is needed. Specifically, additional tools and capabilities are needed to remove the residual manual process and interfaces. As stated earlier, many of the early CD-ROM electronic catalogues provided enhanced searching capabilities. However, these same catalogues were not integrated into either the buyer's or seller's back-office systems. Thus, the following manual actions were required:

- buyer — print and send the order form by facsimile,
 - manually enter the order information into the buyer's accounting system,
 - identify and correct erroneously entered information,
- seller — retrieve the faxed order form,
 - manually enter the order information into the seller's order entry/fulfilment system,
 - identify and correct erroneously entered information.

In order to achieve a fully functional EC environment, the following is the minimum required:

- full integration between the electronic catalogue and the seller's back-office,
- full integration between the buyer's back-office and the electronic catalogue ordering environment,

more frequent product updates or additions, and may include customer-specific prices or products. As mentioned above, these catalogues tend to use closed, proprietary technology and do not provide an electronic ordering option like EDI-CAT. Rather, the user can select products, create, print, and send an order by facsimile to the vendor.

Dynamic, Internet-based catalogues have recently begun to appear. These catalogues use open, standards-based Web browsers for access. Based on predefined user profiles, these dynamic catalogues may be configured to present customer-specific information such as special products (e.g. private brands), prices, discounts or promotions.

2.5.3 Intelligent agents

What is an intelligent agent? Intelligent agents are much more than the search engines currently found in electronic catalogues. Evolving intelligent agents technology will allow users to concurrently search heterogeneous catalogues, identify products, negotiate pricing and delivery terms, and place orders. Intelligent agents may also identify alternate sources or products.

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In order to achieve a fully functional EC environment, the following is the minimum required:

- full integration between the electronic catalogue and the seller's back-office,
- full integration between the buyer's back-office and the electronic catalogue ordering environment,

- the ability to electronically pass invoice and statement information between the seller's and buyer's back-offices.

4. Today's world

4.1 Who are the players?

The HLSG [1] identified three sets of EC players. Engaged and active players included commercial banks, the ICT industry, payment card operators, public network operators, and retail and distribution segments of the 'business world.' Anticipating and observing players included content providers and the 'general business world'. Lagging participants were identified as governments and consumers.

4.2 US experience and projections

Historically, the United States has been an early adopter of new technologies and trading practices. Not surprisingly, this is very much the case with the Internet and electronic commerce. In November 1996, the International Data Corporation (IDC) published a report [12] predicting unprece-dented growth in the use of the Internet and the World Wide Web on a global scale. Some of this report's predictions included:

- the number of users accessing the Web is forecast to grow to 163 million users by 2000, as compared to 16.1 million users in 1995,
- the global Internet electronic commerce market is expected to generate \$1 billion (US) worth of transactions by 2000,
- business-to-business deals will account for \$667 million (US) worth of transactions by 2000,
- 50% of all Internet users will be located outside the US by 2000, as compared to 30% of users who are outside the US in 1996.

Perhaps the most telling statement in this report related to the accuracy of past reports. In this report, IDC suggested that their previous reports had underestimated the size and growth of the electronic commerce market and this report attempted to correct those understatements.

5. Conclusions

Electronic commerce currently exists in two distinct forms, EDI and Web, and it is clear that these will converge over the next year or two. Specifically, the electronic catalogue, the internal authorisation and submission of orders to the supplier must be closely linked.

As the EDI message sets represent more than a decade of active trading experience and business analysis, it is important that those developing the Web formats take full advantage of this by incorporating the EDI information requirements into their own messages.

References

- 1 High Level Strategy Group for ICT: 'Project on Barriers to Electronic Commerce in Support of SMEs', HLSG Report #2, Edition 1.0 (November 1996).
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Jeffrey Edelheit has worked for BT North America since 1995, and focuses on electronic commerce issues and standards. He was awarded his MSIS in 1990 from George Mason University, Fairfax, Virginia, and his BBA in 1971 from the University of Toledo, Ohio. Prior to joining BTNA, he was a Senior IT Security Engineer for The MITRE Corporation, a US Department of Defense Federally Funded Research and Development Center. While at MITRE, he provided security engineering support for the Joint Chiefs of Staff, several Unified Commands, the National Security Agency, and the

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Michael Miller has worked for BT since 1960, initially in telephone exchange system and software design. He was awarded his PhD in 1969 from the University of Warwick for his work on Digital Network Synchronisation. From the System X project he spent some time in The Netherlands designing digital networks for NATO and subsequently led a team designing software for switching and network management systems. Following a period working on marketing and technology innovation, he has been involved in electronic commerce since 1991. He is currently in Global Engineering

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